

Are Expressions for Music Emotions the Same Across Cultures?

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Abstract

Music evokes profound emotions, yet the universality of emotional descriptors across languages remains debated. A key challenge in cross-cultural research on music emotion is biased stimulus selection and manual curation of taxonomies, predominantly relying on Western music and languages. To address this, we propose a balanced experimental design with nine online experiments in Brazil, the US, and South Korea, involving N=672 participants. First, we sample a balanced set of popular music from these countries. Using an open-ended tagging pipeline, we then gather emotion terms to create culture-specific taxonomies. Finally, using these bottom-up taxonomies, participants rate emotions of each song. This allows us to map emotional similarities within and across cultures. Results show consistency in high arousal, high valence emotions but greater variability in others. Notably, machine translations were often inadequate to capture music-specific meanings. These findings together highlight the need for a domain-sensitive, open-ended, bottom-up emotion elicitation approach to reduce cultural biases in emotion research.

Keywords: Music, Emotion, Culture, Cross-cultural analysis

Introduction

Music uniquely expresses emotions, evokes feelings, and influences our mood, making it a fundamental aspect of human experience (Juslin & Sloboda, 2010; North & Hargreaves, 2008; Juslin & Västfjäll, 2008; Park, Thom, Menickien, Cramer, & Macy, 2019). A central debate in emotion research is whether emotions are innate (Darwin, 1872; Ekman & Friesen, 1971) or shaped by language and culture (Russell, Bachorowski, & Fernández-Dols, 2003; Barrett, Adolphs, Marsella, Martinez, & Pollak, 2019). Universalist theories, such as basic emotion theory (Ekman, 1992), argue that emotions are biologically hardwired and universally recognized, emphasizing a core set of emotions like “happiness,” “sadness,” and “anger.” Supporting this view, studies suggest that basic emotions can be recognized across cultures, even in unfamiliar musical traditions (Balkwill & Thompson, 1999; Fritz et al., 2009). However, other research highlights significant cross-cultural differences in emotional experience (Jackson et al., 2019).

A related question in emotion research is the terminology used to describe emotions. Traditionally, emotion research has focused on a small number of basic emotion categories (Descartes, 1989; Ekman, 1992). Additionally, emotions have been proposed to exist along dimensions such as arousal (emotional activation) and valence (positive to negative; Russell, 1980). However, recent studies suggest that de-

scribing musical emotions requires a richer vocabulary beyond a limited set of terms and is potentially described by a high dimensional space (Cowen, Sauter, Tracy, & Keltner, 2019; Cowen & Keltner, 2021; Eerola & Vuoskoski, 2013; Eerola & Saari, 2025).

Cross-cultural research on music and emotion has traditionally relied on a small number of terms, often adapted from Western theories (Balkwill & Thompson, 1999; Fritz et al., 2009). However, large-scale studies using online data reveal substantial variability in emotional terms across languages worldwide (Jackson et al., 2019), suggesting that the taxonomy of musical emotions may also differ across cultures. Furthermore, mood perception studies show high agreement on basic attributes like “sad” and “cheerful” across cultures, while terms such as “dreamy” or “love” can have rather culture-specific interpretations (Lee, Höger, Schönwiesner, Park, & Jacoby, 2021).

How can we characterize music emotion terms across cultures? Previous research has often relied on a limited and biased selection of musical stimuli (Fritz et al., 2009; Balkwill & Thompson, 1999) and an arbitrarily chosen set of emotion terms (Juslin & Västfjäll, 2008; Juslin & Sloboda, 2010). Additionally, these terms have primarily been derived from English and other Western languages, while the musical materials studied have predominantly come from Western traditions. These limitations greatly hinder our ability to capture the variability of cultural nuances and discuss its universalities (Blasi, Henrich, Adamou, Kemmerer, & Majid, 2022).

To address these limitations, we propose a novel pipeline designed to reduce bias and improve cross-cultural comparisons. To mitigate stimulus bias, we carefully sampled a balanced set of 270 songs from Brazil, South Korea, and the US (Lee et al., 2021; Figure 1A). We then conducted a series of experiments, recruiting participants from these three countries (Figure 1B) who performed tasks in their native language. Crucially, all participants engaged with the same shared music database, including both music from their own country and from the other two. This design avoids an a priori dominance of one culture in the stimulus set.

In each country, a group of participants generated emotion-related tags through an open-ended process (Figure 1C). This allowed participants to freely express relevant emotional terms without being constrained by a predefined taxonomy. The iterative nature of the process also enabled participants

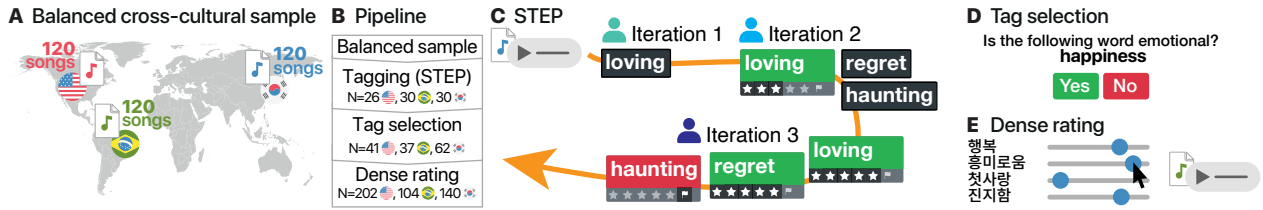


Figure 1: Experimental Setup. **A**: Music collection in three cultures. **B**: Human-in-the-loop pipeline conducted in the following order: **(C)** STEP paradigm to obtain a list of words per culture, **(D)** discrete emotionality rating to select emotional words, and **(E)** dense rating of each stimulus along the select terms.

to refine, rate, and correct each other’s tags, improving the quality of the final term set. By including diverse music from all three cultures, we ensured that a broad emotional spectrum was covered, avoiding biases that could arise from presenting only a narrow range of stimuli (e.g., if only happy songs were presented, terms like “sad” would not emerge).

A separate group of participants then validated whether the collected terms were truly emotion-related (Figure 1D). This step was necessary because participants in the first phase sometimes reported descriptive musical attributes (e.g., “loud”) alongside emotional terms. By establishing a consensus, we differentiated emotion-related descriptors from general musical characteristics.

Since emotion terms may have weak or variable associations with specific stimuli (Toivonen et al., 2012; Durán, Reisenzein, & Fernández-Dols, 2017; Siegel et al., 2018; van Rijn & Larrouy-Maestri, 2023), we accounted for the possibility that multiple terms could apply to a given song (“mixed emotions”). To investigate this, we recruited three additional participant groups—one per culture—who rated the emotions of a subset of songs using the validated taxonomy (Figure 1E). Because all participants listened to the same physical musical stimuli, we could directly compare how emotions were represented across languages.

By analyzing correlations between terms within and across languages, we identified patterns of emotional similarity. Low correlations across songs indicate that participants use these terms in distinct musical contexts, whereas high correlations suggest a stronger semantic connection. We can then use this correlation to map the relationships between emotional terms both within and across cultures.

This pipeline provides a data-driven approach to understanding music emotions cross-culturally, offering new insights into how emotion expressions in music vary across languages and cultures.

Methods

Song Selection

We use a dataset of 360 pop songs originally introduced in (Lee et al., 2021), which evenly samples songs from Brazil, Korea, and the United States. These songs were retrieved from major music charts ensuring a representative selection of popular music trends.

From those 360 songs, 120 songs were presented in each country during the term elicitation phase, consisting of 60 songs from their own culture and 30 songs each from the other two cultures (see ‘Open-ended tagging using STEP’). Each song was presented for 15 seconds in length.

During the rating phase (see ‘Dense rating’), a smaller subset of 60 songs (20 songs each from the three countries) were presented uniformly across all three countries. To maximize the diversity of these songs, they were systematically sampled from the entire pool considering the acoustic features of these songs obtained from Spotify (“danceability”, “energy”, “loudness”, “speechiness”, “acousticness”, “instrumentalness”, “liveness”, and “tempo”), and temporal distribution (two songs per release year for each country).

Participants

We recruited independent sets of participants for each experiment we ran. Recruitment criteria included being at least 18 years old, residing in the target country, and speaking the target language as their primary language. Brazilian and US participants were recruited via Prolific (<https://prolific.com>), while Korean participants were exclusively recruited through CINT (<https://www.cint.com>) due to limited availability on Prolific. Compensation was standardized at £9 per hour for Prolific participants and adjusted to the local minimum wage for CINT participants.

Participants were required to pass a headphone screening task (Milne et al., 2021) before participating in experiments that involved listening to music clips.

In addition, to control for language proficiency, all participants completed a vocabulary test (van Rijn et al., 2023).

Overall we recruited a total of N=672 participants across all experiments.

Procedure

All experiments were conducted using PsyNet¹ (Harrison et al., 2020), an automated recruitment framework for large-scale online studies. Experimental texts were translated using machine translation and then verified by native speakers from each language. Participants provided informed consent

¹PsyNet: <https://psynetdev.gitlab.io/PsyNet/>

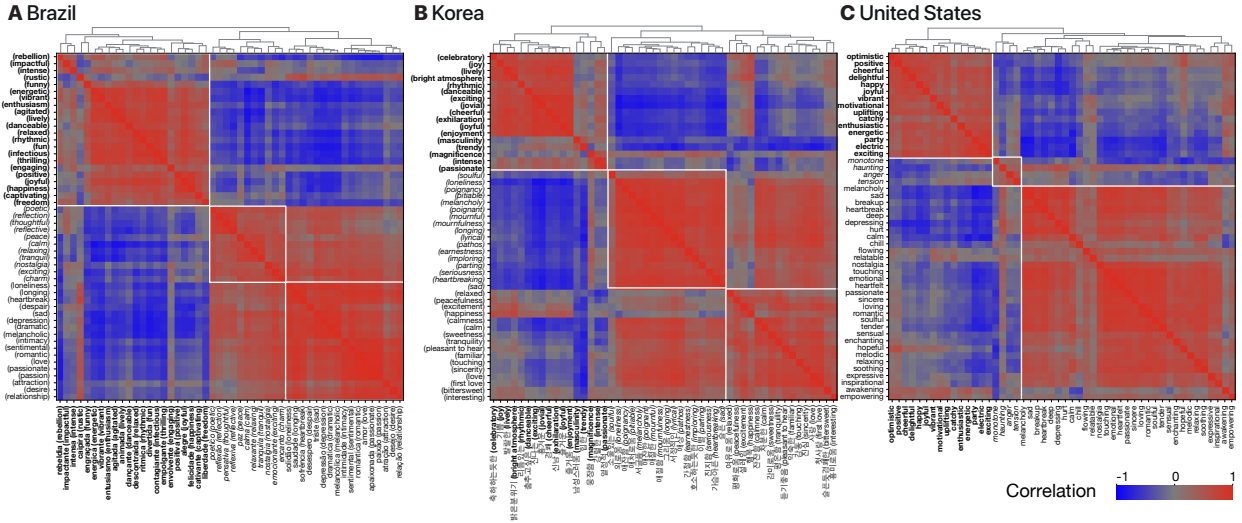


Figure 2: Pearson correlation across 50 emotional terms in Portuguese (Brazil, **A**), Korean (Korea, **B**), and English (US **C**). Correlation matrices are sorted by the order in the dendrogram obtained via agglomerative clustering.

under an approved ethics protocol (Max Planck Ethics Council #202142).

Open-ended tagging using STEP. To capture musical experiences without relying on pre-defined taxonomies, we used STEP (Marjeh et al., 2023; van Rijn, 2024), an open-ended tagging pipeline, where participants generate, evaluate, and refine emotional descriptors over successive iterations (Figure 1C).

In each iteration, (1) participants listened to the full music clip, provided new tags describing their emotional experience, (2) rated existing tags (if previous participants provided tags), and (3) flagged inappropriate ones (e.g., typos, irrelevant responses). They were instructed to use single-word tags that reflect both musical and emotional characteristics, avoiding genre labels and direct transcription of lyrics. Tags were written in their native language, with a 15-character limit, and multiple entries were allowed. Each participant annotated 15 music clips to balance the workload and maintain diversity.

We recruited three groups of participants (US=26, Brazil=30, Korea=30) to annotate 120 randomly selected songs. For each song, after five iterations, this process produced a weighted bag-of-words representation for each song. By analyzing tag co-occurrences, we derived an emergent taxonomy of emotions. This approach is particularly effective for cross-cultural studies, as it iteratively builds culturally grounded semantic representations.

Tag selection. STEP process in each country resulted in 158, 194, and 299 labels for the US, Brazil, and Korea respectively. To create culturally relevant emotional descriptors, we followed a two-step process: cleaning and validation.

In the first step, we cleaned the tags using an automated pipeline (specifically: lemmatizing words, and correcting ty-

pos with fuzzy matching, see Niedermann et al. 2024 for more details). Tags containing invalid characters or formats were excluded. After automated cleaning, words with minor spelling variations were manually merged into identical tags. This process resulted in a final list of 150, 186, and 259 emotional descriptors in the US, Brazil, and Korea, respectively (Figure 1D).

To subselect tags referring explicitly to emotional experiences, we conducted a tag selection experiment (US=41, Brazil=37, Korea=62 participants). Each tag was shown to the participants and asked whether the given tag could be used to describe emotions in music or not. An average of 12 emotionality ratings were collected for tags.

Based on this data, we computed the mean emotionality per tag. Labels that met the 50% agreement threshold were considered valid emotional terms within that culture. We subselected the top 50 most emotional terms in each culture. The final set of 50 terms per language and culture served as the emotional taxonomy for the next step. This number was chosen based on prior studies suggesting that around 50 emotions effectively capture a significant portion of emotional variability (Cowen, Sauter, et al., 2019).

Dense rating. While participants in STEP provide ratings for the relevance of each of the tags in the stimuli, this rating is fairly sparse (i.e., each stimulus receives only a small number of the most appropriate tags). Since emotions can capture more nuanced aspects of music, it is important to account for weakly relevant annotations to better reflect the complexity of emotional responses.

Thus, for a subset of songs (see ‘Song Selection’), we conducted a dense rating experiment (US=202, Brazil=104, Korea=140 participants) to study the associations across emotion terms for music within and across cultures, (Figure 1E).